

Scenario 1: Andalia

Andalia is a landlocked middle income country with a very large population and high mobile phone penetration rates. The NSO conducted a census three years ago, and regularly collects administrative and household surveys to produce statistics on topics such as household income and expenditure, domestic tourism and travel, labour market, etc. The DG of Andalia National Statistical Office is keen to combine this data with new data sources, including using the emerging methodologies for analysing MNO call detail records (CDRs) to produce updated intercensal population statistics which also take into account issues such as rural-urban migration. In addition, the disaster management agency of Andalia would like to use MPD to support its disaster response mandate. The status of taking these ideas forward is as follows:

- Talks with the CEO at the biggest telecoms service provider indicate there could be an opportunity for sharing some data, especially if the NSO is willing to also share population data from the census at a more granular level than is currently available in order to improve its own business intelligence. The other, smaller MNOs are less enthusiastic but may be persuaded if they can generate revenue from the sale of new data products, either to the NSO or for sale to private sector purchasers.
- Andalia has a data protection law and strong data privacy regulations. The data protection commissioner is new, well-respected and has shown an interest in the idea but has serious concerns about protecting individual privacy.
- The Ministry of Telecommunications regulates MNOs and issues their licenses. It is already using MPD for compliance purposes and has trialled using the data for measuring Information Society statistics such as % of population with access to the internet. After a request by the DG of the NSO, a letter of non-objection has been received from the Ministry for using MNO data for official statistics on migration.
- Andalia's NSO is under-staffed and overstretched. Some data scientists recently joined and are working on geospatial data but have no experience of using MPD.
- Internet costs are high in Andalia, and IT equipment is expensive. Importing servers for a recent IT improvement project has been a headache.

A) Identifying the relevant incentives to motivate data sharing with NSOs

Which of the following are existing or potential incentives for MPD data sharing in this scenario?

Incentives / Motives for sharing MPD with NSO for statistics	
Compliance with regulation / legal mandate	
Public Recognition / Corporate Social Responsibility (CSR)	
Data Analysis Capacity Building	
New Products & Monetization	X
Data Reciprocity	X
Innovation and R&D	X

B) Key inputs and necessary pre-conditions required for an effective MPD initiative

Does the country have in place the following key ingredients for producing statistics from MPD?

Critical inputs and pre-conditions	Yes	No	Don't know
A clear specification of which data is to be shared and for what purpose	x		
Regulatory approval from the relevant bodies, as applicable (Telecoms regulator, Data Protection Authority)	X - TRA	X - DPA	
A signed data sharing agreement describing how data will be shared, who holds the data and who will receive it		X	
IT infrastructure for storing and processing the mobile phone data		X	
Staff with the necessary expertise to collect and process MPD		X	

Note: TRA = Telecoms Regulatory Authority; DPA = Data Protection Authority

C) Assessing the feasibility of an MPD initiative

The following are key dimensions for assessing the feasibility of an MPD initiative. From what you know of the current situation, how would you describe each dimension?

Key dimensions	Enabling	Disabling	Don't know / more info required
Suitability of legal and governance environment	Potentially but...		...X
Stakeholder engagement and alignment	Potentially but...		...X
Suitability of data infrastructure		At present but...	...X
Suitability of data environment	X		
Human resource capacity and availability		X	

Scenario 2: Zampetiland

Zampetiland is an archipelago nation. It relies heavily on domestic and international tourism. The Ministry of Tourism recently did a pilot project testing the use of MPD for targeted marketing purposes. They used the data to produce tourism statistics and shared these with local municipalities who hope to increase tourism revenue. The results were encouraging. The NSO would like to scale the work up to produce national-level statistics on domestic and international tourism. Another application the NSO would like to explore is using MPD for producing transport-related statistics. In addition:

- Regulatory approval for the sharing of MNO data was obtained previously from relevant regulatory bodies for the pilot. However, there have been staff changes since then. A signed agreement remains in place for sharing of mobile phone data between the MNO involved in the pilot and the Ministry of Tourism, but not yet with the NSO. The agreement will expire at the end of the year.
- The pilot project relied on staff at the MNO running the code to process individual records into aggregated datasets for sharing with the Ministry of Tourism. It's unclear at this time if the MNO is still willing to assign their staff to this role, especially if the work and applications significantly increase. The MNO has also recently recruited a new cohort of data scientists keen to progress their careers and recently showed an interest in how a collaboration could support its brand recognition.
- The NSO has been active in sourcing data from surveys, census and administrative data, but in-person data collection is very expensive because of the country's many islands.

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Compliance with regulation / legal mandate	
Public Recognition / Corporate Social Responsibility (CSR)	X
Data Analysis Capacity Building	X
New Products & Monetization	
Data Reciprocity	
Innovation and R&D	X

B) Key inputs and necessary pre-conditions required for an effective MPD initiative

Does the country have in place the following key ingredients for producing statistics from MPD?

Critical inputs and pre-conditions	Yes	No	Don't know
A clear specification of which data is to be shared and for what purpose	X		
Regulatory approval from the relevant bodies, as applicable (Telecoms regulator, Data Protection Authority)	X - pilot	X - scale-up	
A signed data sharing agreement describing how data will be shared, who holds the data and who will receive it	X - <u>but</u> expires soon	X - not for NSO	
IT infrastructure for storing and processing the mobile phone data			X
Staff with the necessary expertise to collect and process MPD	X - MNO	X - NSO	

C) Assessing the feasibility of an MPD initiative

The following are key dimensions for assessing the feasibility of an MPD initiative. From what you know of the current situation, how would you describe each dimension?

Key dimensions	Enabling	Disabling	Don't know / more info required
Suitability of legal and governance environment	X		
Stakeholder engagement and alignment	X - pilot		X - scale-up
Suitability of data infrastructure			X
Suitability of data environment	X		
Human resource capacity and availability	Capacity at MNO but...		...availability?

Scenario 3: Banjetiga

During the COVID-19 pandemic, The Republic of Banjetiga piloted a social protection programme that combined geospatial data and MNO data to identify vulnerable and affected populations and send them financial support through mobile money providers. The NSO and the Ministry of Women, Children and Social Welfare are keen to expand the programme. However, the pilot was conducted largely with support from academics at an overseas university who now work on other topics and in any case the country's politicians have recently been emphasising the importance of data sovereignty and so will not support external actors analysing the data. The NSO is now exploring its options. In addition:

- A new National Data Strategy is being developed and will soon be out for consultation. It includes, inter alia, a provision mandating the sharing of certain types of privately-held data for statistical purposes and a provision for creating a new, stand-alone Data Protection Authority.
- The National Communications Council, which regulates the telecoms industry, is positive about the use of this data by the NSO and government ministries.
- The MNOs that participated in the pilot are interested in scale-up and willing to process their data because it uses their mobile money services for large-scale distribution of public funds. Each MNO can also see the potential for building a positive public image if they are involved; but they are all simultaneously concerned about the sharing of commercially sensitive data with any competitors.
- A large-scale digital transformation project has significantly increased the cellphone coverage in the country, and several tech companies have submitted bids to build high capacity data centres in the country which is rich in renewable energy resources.

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Public Recognition / Corporate Social Responsibility (CSR)	X
Data Analysis Capacity Building	
New Products & Monetization	X
Data Reciprocity	
Innovation and R&D	